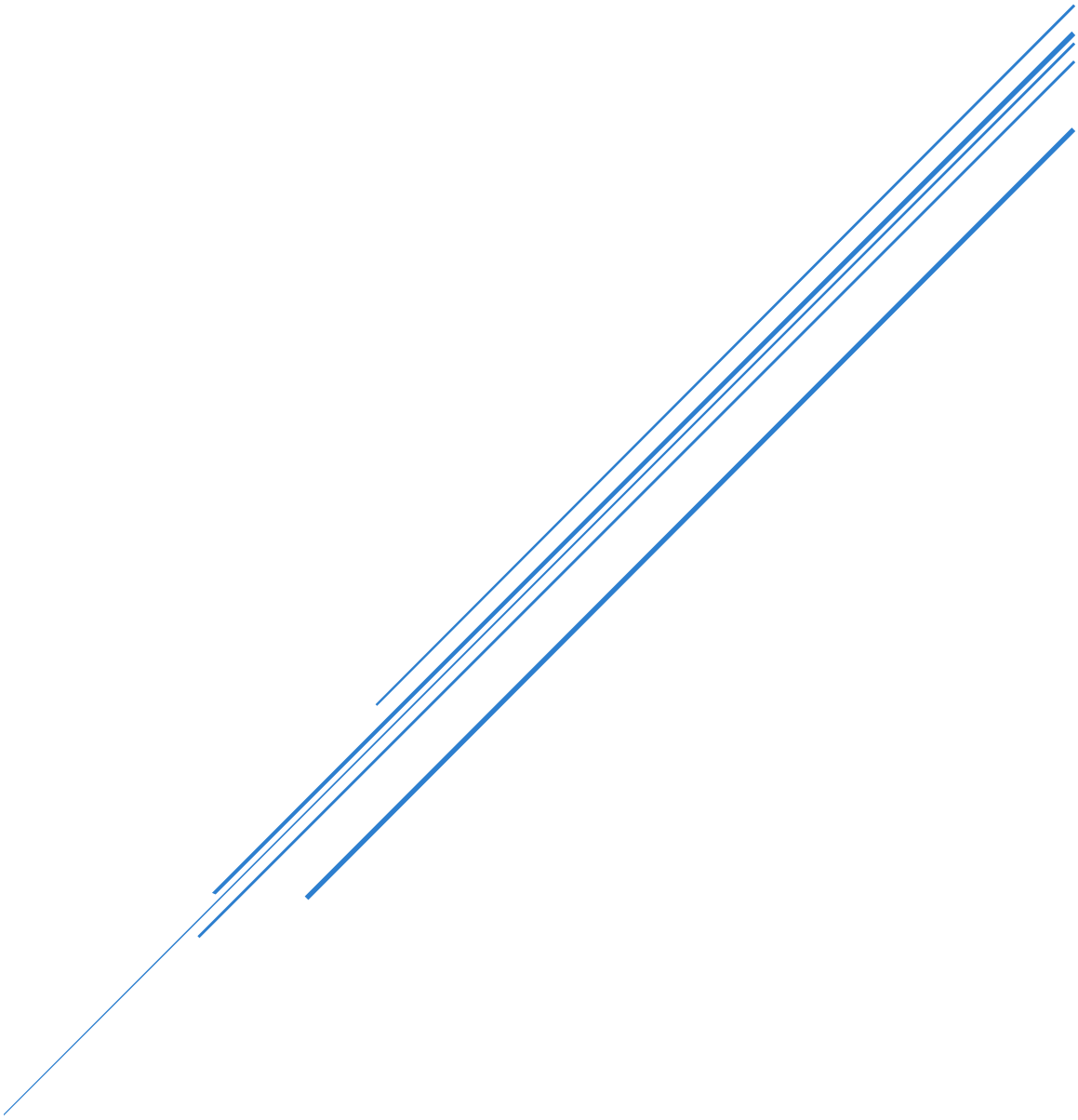
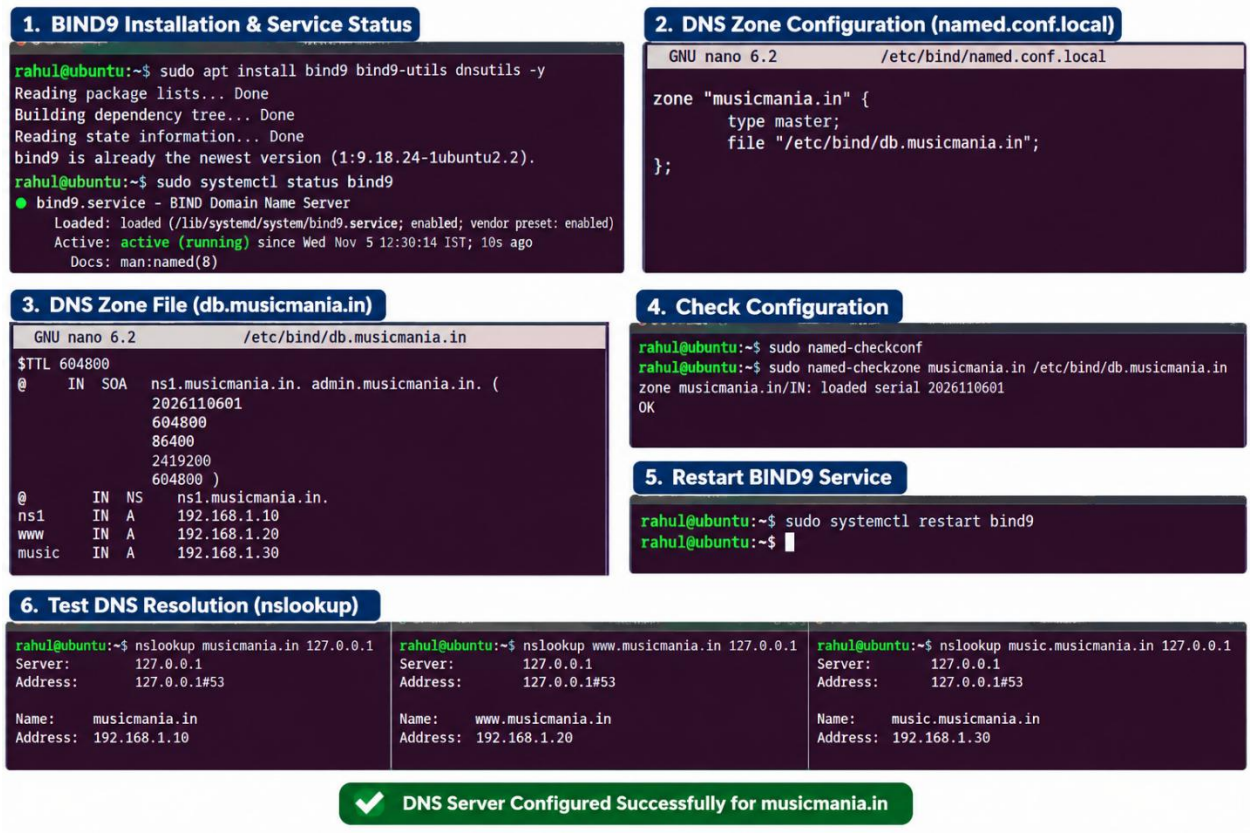
# DAY - 3 DNS SERVER CONNFIGURATION

Assignment - 11



1. **Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.**

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**I configured a DNS server using BIND9 on Ubuntu Linux. First, I installed BIND9 and checked its service status to ensure that it was running properly.**

**Next, I created a forward lookup DNS zone named musicmania.in in the named.conf.local file. I created a zone file named db.musicmania.in and added DNS records for the domain, including NS and A records for ns1, www, and music.**

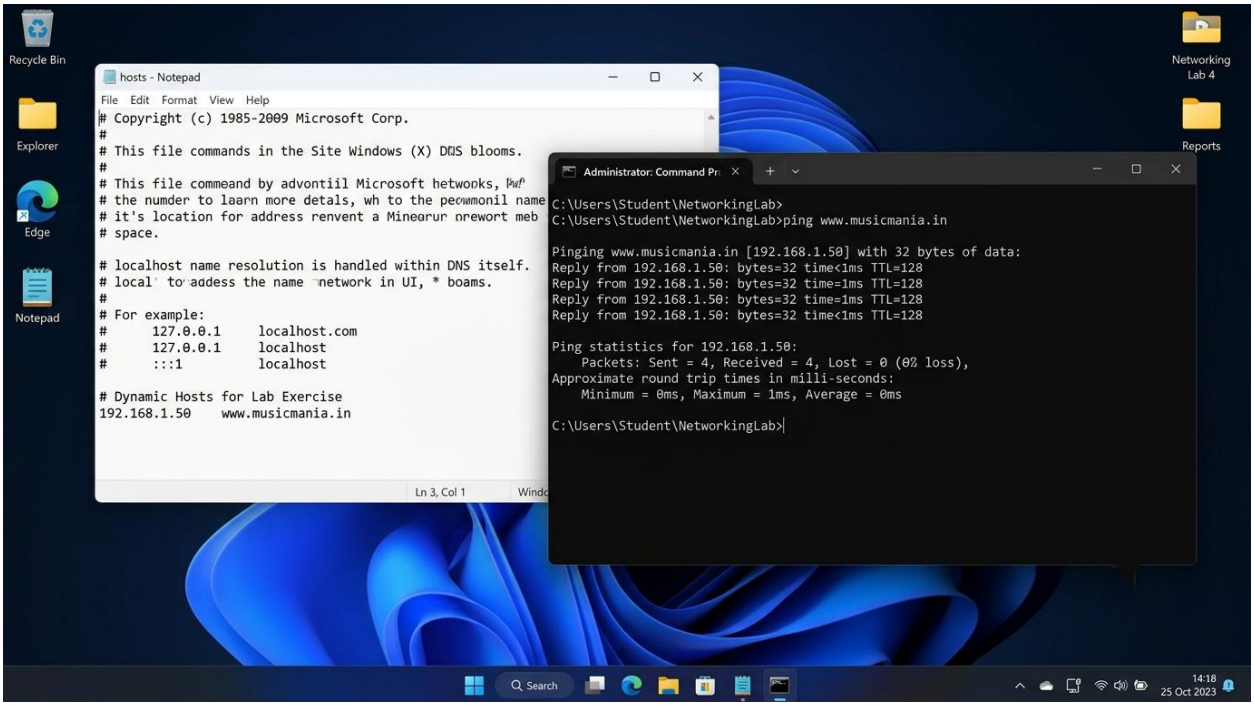
**After that, I verified the configuration using named-checkconf and named-checkzone commands and restarted the BIND9 service.**

**Finally, I tested DNS resolution using the nslookup command for musicmania.in, www.musicmania.in, and music.musicmania.in.**

**Result: The BIND9 DNS server was configured successfully, and the DNS zone musicmania.in was created with the required DNS records.**

2. **Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.**

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1. **Notepad: Hosts file contains the mapping: 192.168.1.50 www.musicmania.in**
2. **Command Prompt: Ping command executed: ping www.musicmania.in**
3. **Successful resolution: The domain resolves to 192.168.1.50.**
4. **Successful replies: 4 packets sent, 4 received, 0% packet loss.**

**Answer for Assignment**

**Procedure:**

1. **Open Notepad as Administrator.**
2. **Open the Windows Hosts file from C:\Windows\System32\drivers\etc\hosts.**
3. **Add the following entry:**

**192.168.1.50 www.musicmania.in**

4. **Save the file.**
5. **Open Command Prompt as Administrator.**
6. **Execute the following command: ipconfig /flushdns**
7. **Verify the mapping using:**

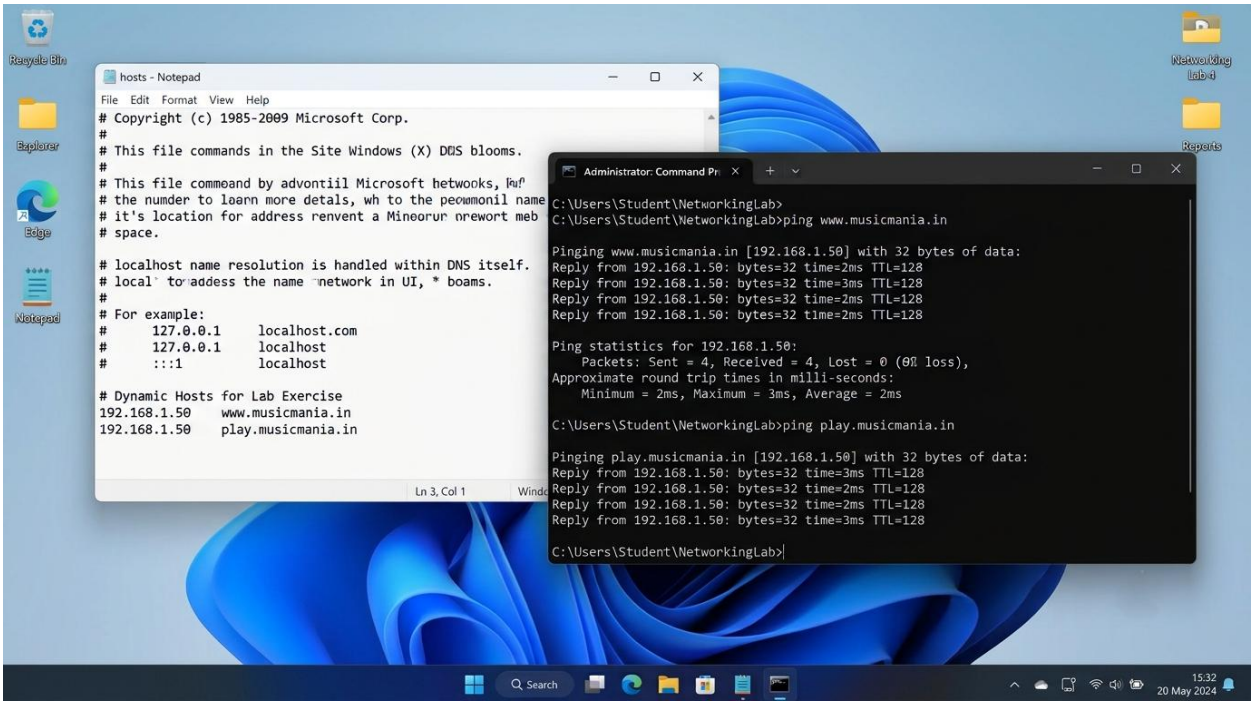
**ping www.musicmania.in**

**Observation: The command successfully resolved www.musicmania.in to 192.168.1.50. All four packets were received with 0% packet loss.**

**Result: Successfully configured and verified the local hostname mapping of www.musicmania.in to IP address 192.168.1.50 using the Windows Hosts file.**

3. **Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.**

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**Procedure:**

1. **Open Notepad as Administrator.**
2. **Open the Windows Hosts file from C:\Windows\System32\drivers\etc\hosts.**
3. **Add the following entries:**

**192.168.1.50 www.musicmania.in**

**192.168.1.50 play.musicmania.in**

4. **Save the Hosts file.**
5. **Open Command Prompt as Administrator.**
6. **Flush the DNS cache using:**

**ipconfig /flushdns**

**7. Test the first domain:**

**ping www.musicmania.in**

**8. Test the second domain:**

**ping play.musicmania.in**

**Observation:**

**As shown in the image:**

- **www.musicmania.in successfully resolves to 192.168.1.50.**
- **play.musicmania.in also successfully resolves to 192.168.1.50.**
- **Both domains receive successful ping replies.**
- **The IP address is the same for both domain names.**

**Result:**

**Successfully simulated a CNAME-style alias for play.musicmania.in pointing to www.musicmania.in. Both domain names resolved to the same IP address, 192.168.1.50, and connectivity was verified using the ping command.**

4. **Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.**
   **Hint: Include at least three different troubleshooting commands or checks in your documentation.**

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**Step-by-Step Troubleshooting Process**

**Step 1: Check DNS Configuration**

- **Open the DNS configuration or Hosts file.**
- **Verify that the domain name and IP address are entered correctly.**
- **Check for spelling mistakes and incorrect IP addresses.**

**Example entry:**

**192.168.1.50 musicmania.in**

**Step 2: Check Network Connectivity**

**Open Command Prompt and execute:**

**ping 8.8.8.8**

**This checks whether the computer has internet connectivity.**

**Step 3: Flush DNS Cache**

**Run the following command:**

**ipconfig /flushdns**

**This clears old DNS records stored on the computer.**

**Step 4: Restart DNS Service**

**On Windows Server, open Command Prompt as Administrator and execute:**

**net stop dns**

**net start dns**

**This restarts the DNS Server service.**

**Step 5: Test DNS Resolution**

**Use the following command:**

**nslookup musicmania.in**

**This checks whether the domain resolves to the correct IP address.**

**Step 6: Check DNS Server Configuration**

**Run:**

**ipconfig /all**

**Check whether the correct DNS server IP address is configured on the computer.**

**Step 7: Verify the Result**

**Run the DNS lookup again:**

**nslookup musicmania.in**

**If the correct IP address is displayed, the DNS resolution problem has been fixed.**

**Troubleshooting Commands Summary**

| Command | Purpose |
|---|---|
| ping 8.8.8.8 | Checks network connectivity |
| ipconfig /flushdns | Clears DNS cache |
| ipconfig /all | Displays network and DNS settings |
| nslookup musicmania.in | Tests DNS resolution |
| net stop dns | Stops DNS service |
| net start dns | Starts DNS service |

**Result**

**Successfully documented the step-by-step DNS troubleshooting process. Checked DNS configuration, network connectivity, DNS cache, and DNS service status. Used commands such as ping, ipconfig, and nslookup to identify and resolve DNS resolution issues.**